

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 94 on 04/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMDQwODIzMzhfMzA2MTg%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

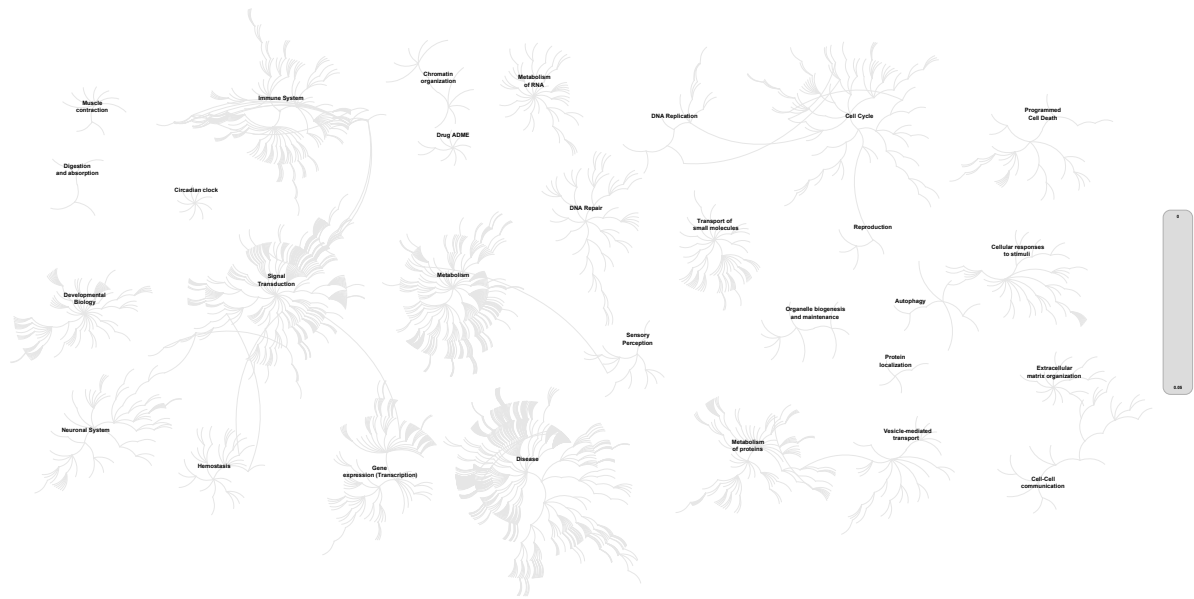
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [🔗](#)
- 127 out of 192 identifiers in the sample were found in Reactome, where 700 pathways were hit by at least one of them.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'UniProt'.
- The unique ID for this analysis (token) is MjAyNTEyMDQwODIzMzhfMzA2MTg%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 7 most relevant pathways sorted by p-value.

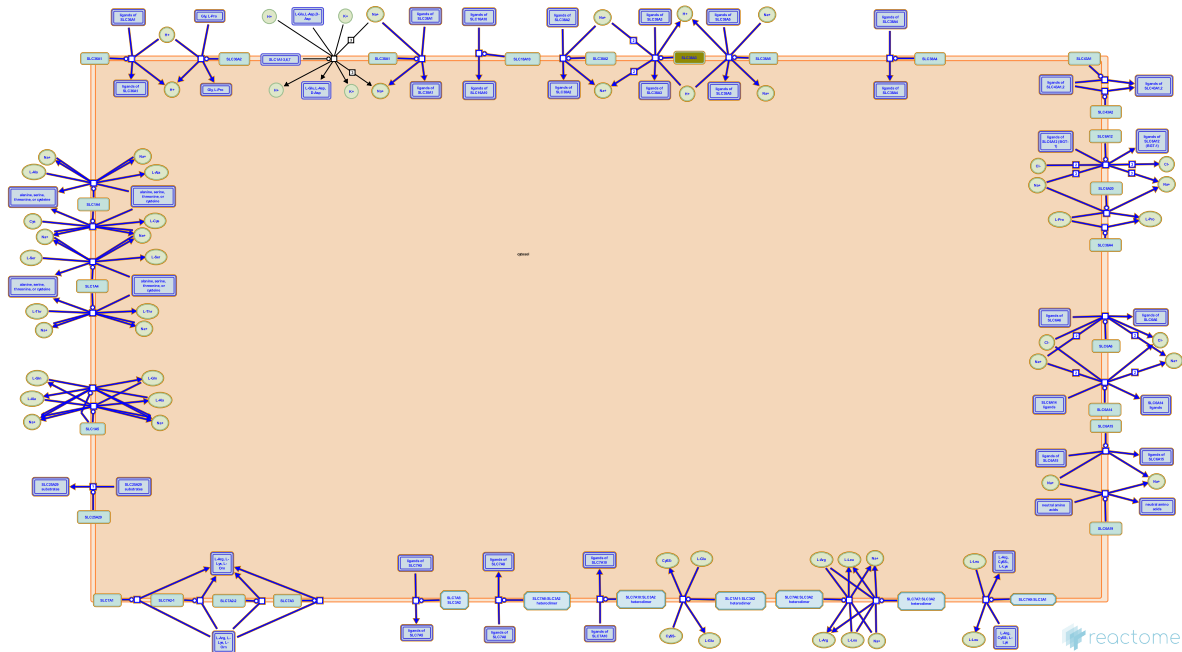
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Amino acid transport across the plasma membrane	1 / 33	0.003	0.412	0.981	1 / 35	0.002
SLC-mediated transport of amino acids	1 / 38	0.003	0.457	0.981	1 / 36	0.002
Post-translational modification: synthesis of GPI-anchored proteins	1 / 93	0.008	0.777	0.981	1 / 16	0.001
SLC-mediated transmembrane transport	1 / 244	0.02	0.981	0.981	1 / 193	0.012
Transport of small molecules	1 / 723	0.06	1	1	1 / 462	0.029
Post-translational protein modification	1 / 1,483	0.123	1	1	1 / 606	0.039
Metabolism of proteins	1 / 2,113	0.175	1	1	1 / 929	0.059

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Amino acid transport across the plasma membrane (R-HSA-352230)



Cellular compartments: plasma membrane.

Amino acid transport across plasma membranes is critical to the uptake of these molecules from the gut, to their reabsorption in the kidney proximal tubules, and to their distribution to cells in which they are required for the synthesis of proteins and of amino acid derived small molecules such as neurotransmitters. Physiological studies have defined 18 "systems" that mediate amino acid transport, each characterized by its amino acid substrates, as well as its pH sensitivity and its association (or not) with ion transport. More recently, molecular cloning studies have allowed the identification of the plasma membrane transport proteins that mediate these reactions. Amino acid uptake mediated by 17 of these transporters is annotated here (Broer 2008).

References

Broer S (2008). Amino acid transport across mammalian intestinal and renal epithelia. *Physiol Rev*, 88, 249-86. [🔗](#)

Edit history

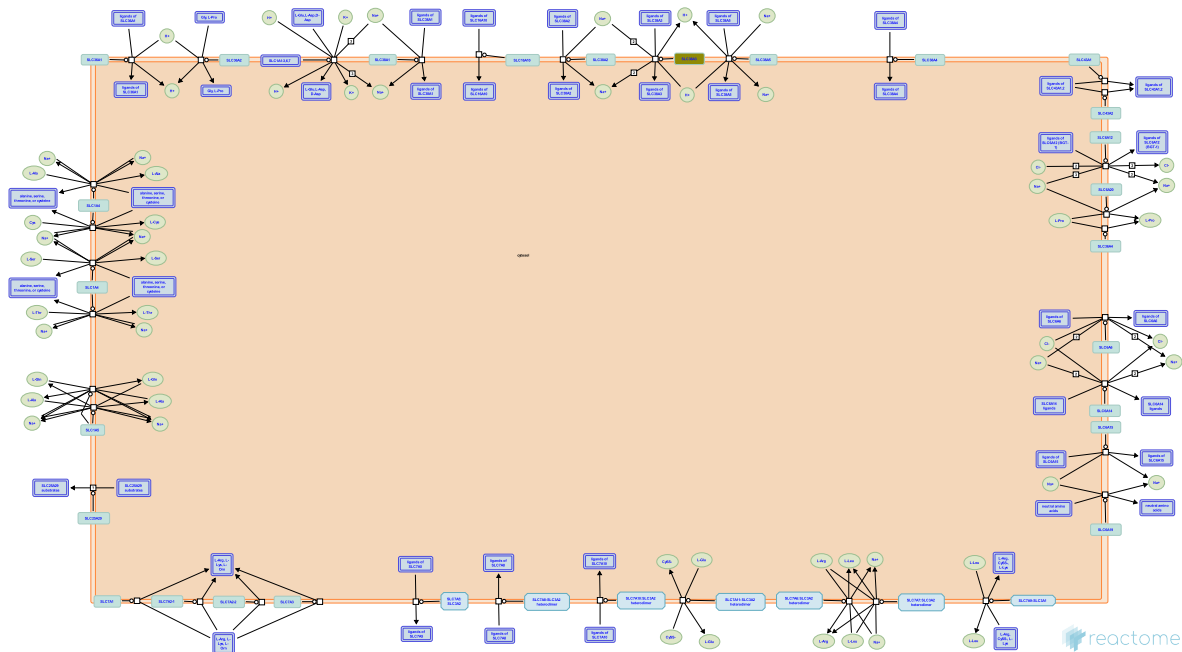
Date	Action	Author
2008-06-02	Created	D'Eustachio P
2008-06-03	Edited	D'Eustachio P
2008-06-03	Reviewed	Jassal B
2008-06-03	Authored	D'Eustachio P
2025-08-12	Reviewed	D'Eustachio P

Date	Action	Author
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P11798	Q99624

2. SLC-mediated transport of amino acids (R-HSA-9958863)



This pathway serves as collection of reactions categorized as SLC-mediated transport of amino acids.

References

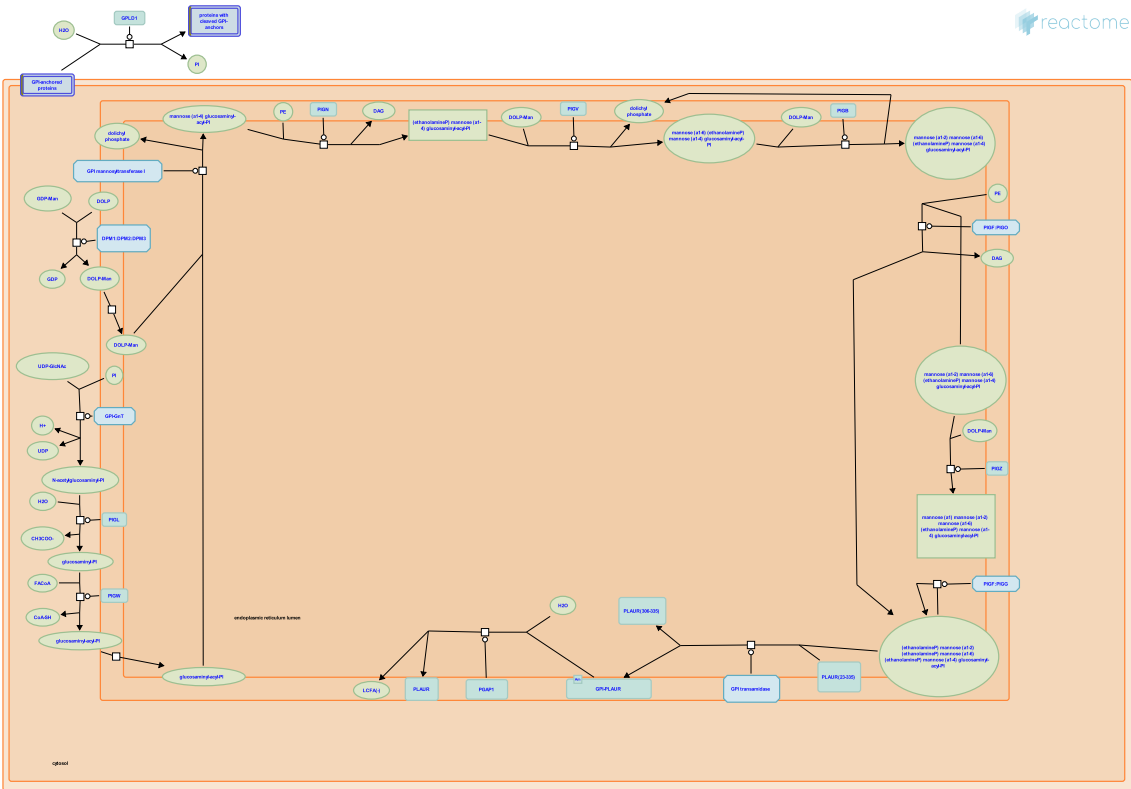
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Date	Action	Author
2025-06-25	Authored	Stephan R
2025-06-25	Created	Stephan R
2025-08-12	Reviewed	D'Eustachio P
2025-08-13	Edited	Stephan R
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P11798	Q99624

3. Post-translational modification: synthesis of GPI-anchored proteins (R-HSA-163125)



Glycosylphosphatidyl inositol (GPI) acts as a membrane anchor for many cell surface proteins. GPI is synthesized in the endoplasmic reticulum. In humans, a single pathway consisting of eleven reactions appears to be responsible for the synthesis of the major GPI species involved in membrane protein anchoring.

As a nascent protein fated to become GPI-anchored moves into the lumen of the endoplasmic reticulum, it is attacked by a transamidase complex that cleaves it near its carboxy terminus and attaches an acylated GPI moiety. The GPI moiety is deacylated, yielding a protein-GPI conjugate that can be efficiently transported to the Golgi apparatus.

References

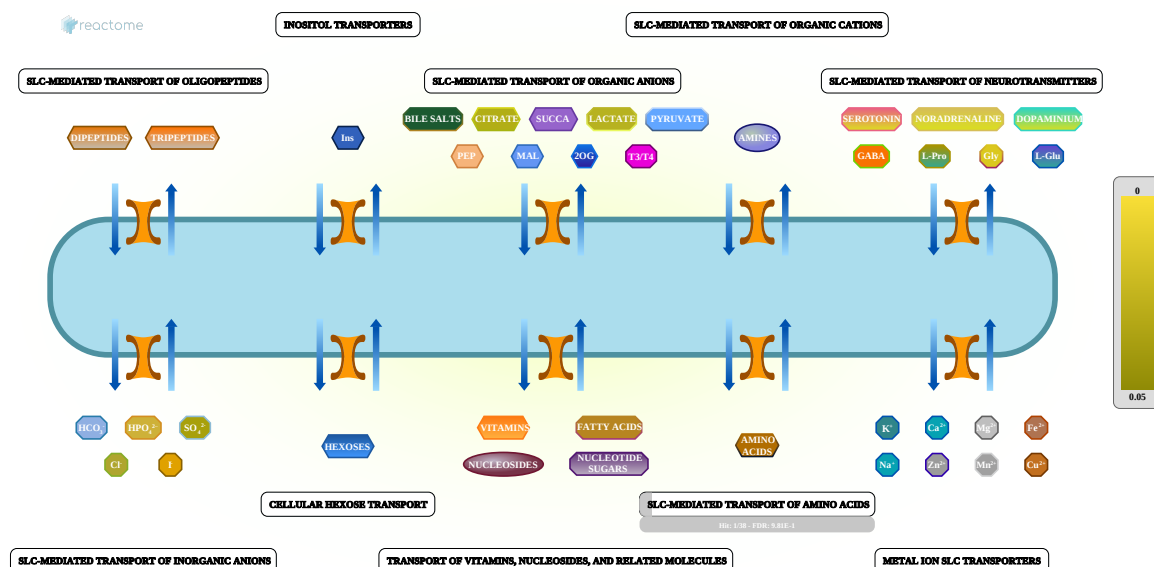
Edit history

Date	Action	Author
2005-04-18	Created	D'Eustachio P
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P26645	Q93070

4. SLC-mediated transmembrane transport ([R-HSA-425407](#))



Proteins with transporting functions can be roughly classified into 3 categories: ATP-powered pumps, ion channels, and transporters. Pumps utilize the energy released by ATP hydrolysis to power the movement of the substrates across the membrane, against their electrochemical gradient. Channels at the open state can transfer the substrates (ions or water) down their electrochemical gradient, at an extremely high efficiency (up to 10⁸ s⁻¹). Transporters facilitate the movement of a specific substrate either against or following their concentration gradient, at a lower speed (about 10² -10⁴ s⁻¹); as generally believed, conformational change of the transporter protein is involved in the transfer process.

According to the Human Genome Organization (HUGO) Gene Nomenclature Committee, all human transporters can be grouped into the solute-carrier (SLC) superfamily (<http://www.genenames.org/genefamilies/SLC>). Currently, there are 55 SLC families in the superfamily, with a total of at least 362 putatively functional protein-coding genes (Hediger et al. 2004, He et al. 2009; <http://www.bioparadigms.org/slc/intro.htm>). At least 20-25% amino-acid sequence identity is shared by members belonging to the same SLC family. No homology is shared between different SLC families. While the HUGO nomenclature system by definition only includes human genes, the nomenclature system has been informally extended to include rodent species through the use of lower cases letters (e.g., Slc1a1 denotes the rodent ortholog of the human SLC1A1 gene). And it's worthwhile to mention that pumps, channels and aquaporins are not included in SLC superfamily.

To date, nine SLC gene families (SLC4, SLC5, SLC8, SLC9, SLC12, SLC20, SLC24, SLC26 and SLC34) comprise the group that exclusively transports inorganic cations and anions across membranes. A further eight SLC gene families (SLC1, SLC6, SLC7, SLC16, SLC25, SLC36, SLC38 and SLC43) are involved in the transport of amino acids and oligopeptides (He et al. 2009). Two gene families are responsible for glucose transport in humans. SLC2 (encoding GLUTs) and SLC5 (encoding SGLTs) families mediate glucose absorption in the small intestine, glucose reabsorption in the kidney, glucose uptake by the brain across the blood-brain barrier and glucose release by all cells in the body (Wood & Trayhurn 2003).

SLC transporters are able to transport bile salts, organic acids, metal ions and amine compounds. Myo-Inositol is a precursor to phosphatidylinositols (PtdIns) and to the inositol phosphates (IP), which serve as second messengers and also act as key regulators of many cell functions (Schneider 2015). Mono-, di- and tri-carboxylate transporters mediate the transport of these acids across cellular membranes (Pajor 2006, Morris & Felmlee 2008). Essential metals are transported by metal-transporting proteins, which also control their efflux to avoid toxic build-up (Bressler et al. 2007). The SLC6 gene family encodes proteins that mediate neurotransmitter uptake in the central nervous system (CSN) and peripheral nervous system (PNS), thus terminating a synaptic signal (Chen et al. 2004). Urea transport is particularly important in the process of urinary concentration and for rapid urea equilibrium in non-renal tissues (Olives et al. 1994). Choline uptake is the rate-limiting step in the synthesis of the neurotransmitter acetylcholine. SLC genes SLC5A7 and the SLC44 family encode choline transporters (Traiffort et al. 2005). The SLC22 gene family of solute carriers function as organic cation transporters (OCTs), cation/zwitterion transporters (OCTNs) and organic anion transporters (OATs). They play important roles in drug absorption and excretion. Substrates include xenobiotics, drugs, and endogenous amine compounds (Koepsell & Endou 2004).

The human SLC5A6 encodes the Na⁺-dependent multivitamin transporter SMVT (Prasad et al. 1999). SMVT co-transporters biotin (vitamin B7), D-Pantoic acid (vitamin B5) and lipoic acid into cells with Na⁺ ions electrogenically. Four SLC gene families encode transporters that play key roles in nucleoside and nucleobase uptake for salvage pathways of nucleotide synthesis, and in the cellular uptake of nucleoside analogues used in the treatment of cancers and viral diseases (He et al. 2009). The human gene SLC33A1 encodes acetyl-CoA transporter AT1 (Kanamori et al. 1997). Acetyl-CoA is transported to the lumen of the Golgi apparatus, where it serves as the substrate of acetyltransferases that O-acetylates sialyl residues of gangliosides and glycoproteins. Nucleotide sugars are used as sugar donors by glycosyltransferases to create the sugar chains for glycoconjugates such as glycoproteins, polysaccharides and glycolipids. The human solute carrier family SLC35 encode nucleotide sugar transporters (NSTs), localised on Golgi and ER membranes, which can mediate the antiport of nucleotide sugars in exchange for the corresponding nucleoside monophosphates (eg. UMP for UDP-sugars) (Handford et al. 2006). Long chain fatty acids (LCFAs) can be used for energy sources and steroid hormone synthesis and regulate many cellular processes such as inflammation, blood pressure, the clotting process, blood lipid levels and the immune response. The SLC27A family encode fatty acid transporter proteins (FATPs) (Anderson & Stahl 2013). The SLC gene family members SLCO1 SLCO2 and SLCO3 encode organic anion transporting polypeptides (OATPs). OATPs are membrane transport proteins that mediate the sodium-independent transport of a wide range of amphipathic organic compounds including bile salts, steroid conjugates, thyroid hormones, anionic oligopeptides and numerous drugs (Hagenbuch & Meier 2004).

References

- He L, Vasiliou K & Nebert DW (2009). Analysis and update of the human solute carrier (SLC) gene superfamily. *Hum Genomics*, 3, 195-206. [🔗](#)
- Hediger MA, Romero MF, Peng JB, Rolfs A, Takanaga H & Bruford EA (2004). The ABCs of solute carriers: physiological, pathological and therapeutic implications of human membrane transport proteinsIntroduction. *Pflugers Arch*, 447, 465-8. [🔗](#)
- Wood IS & Trayhurn P (2003). Glucose transporters (GLUT and SGLT): expanded families of sugar transport proteins. *Br J Nutr*, 89, 3-9. [🔗](#)
- Schneider S (2015). Inositol transport proteins. *FEBS Lett.*, 589, 1049-58. [🔗](#)

Morris ME & Felmllee MA (2008). Overview of the proton-coupled MCT (SLC16A) family of transporters: characterization, function and role in the transport of the drug of abuse gamma-hydroxybutyric acid. AAPS J, 10, 311-21. [🔗](#)

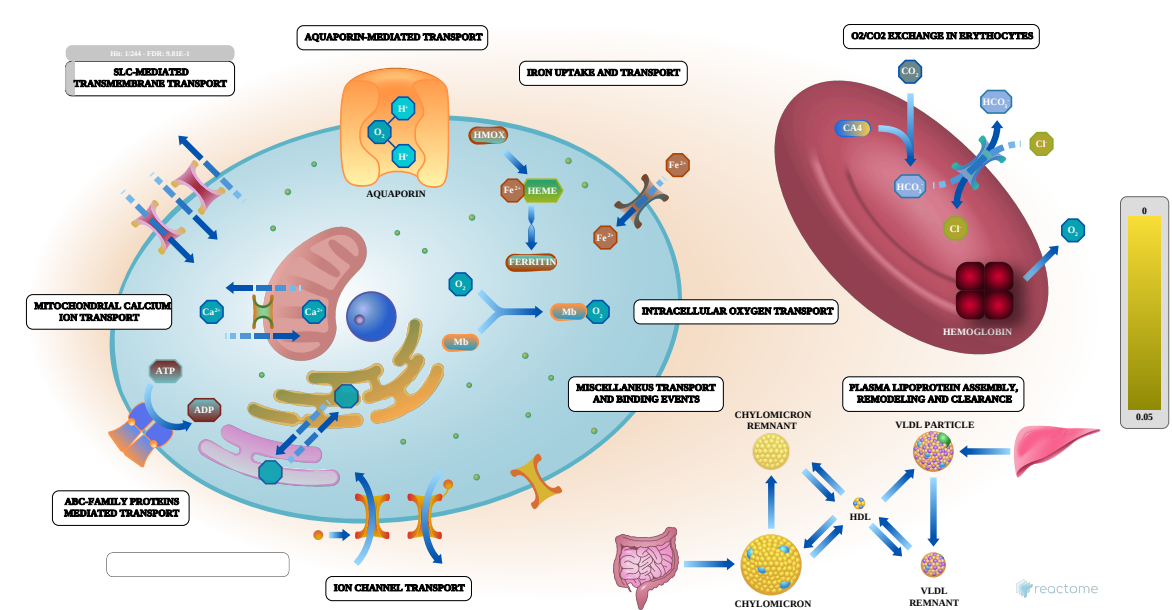
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Date	Action	Author
2009-06-04	Edited	Jassal B
2009-06-04	Authored	Jassal B
2009-06-04	Created	Jassal B
2009-08-24	Reviewed	He L
2014-06-09	Revised	Jassal B
2025-08-12	Reviewed	D'Eustachio P
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P11798	Q99624

5. Transport of small molecules (R-HSA-382551)



By definition cells have a critical separation between inner (cytoplasmic) and outer (extracellular) compartments. This separation provides for protection, gradient assembly, and environmental control but at the same time isolates the interior compartments of the cell from energy resources, oxygen, and raw materials. Cells have evolved a myriad of mechanisms to regulate, and enable transportation of small molecules across plasma membranes and between cellular organelle compartments within cells.

References

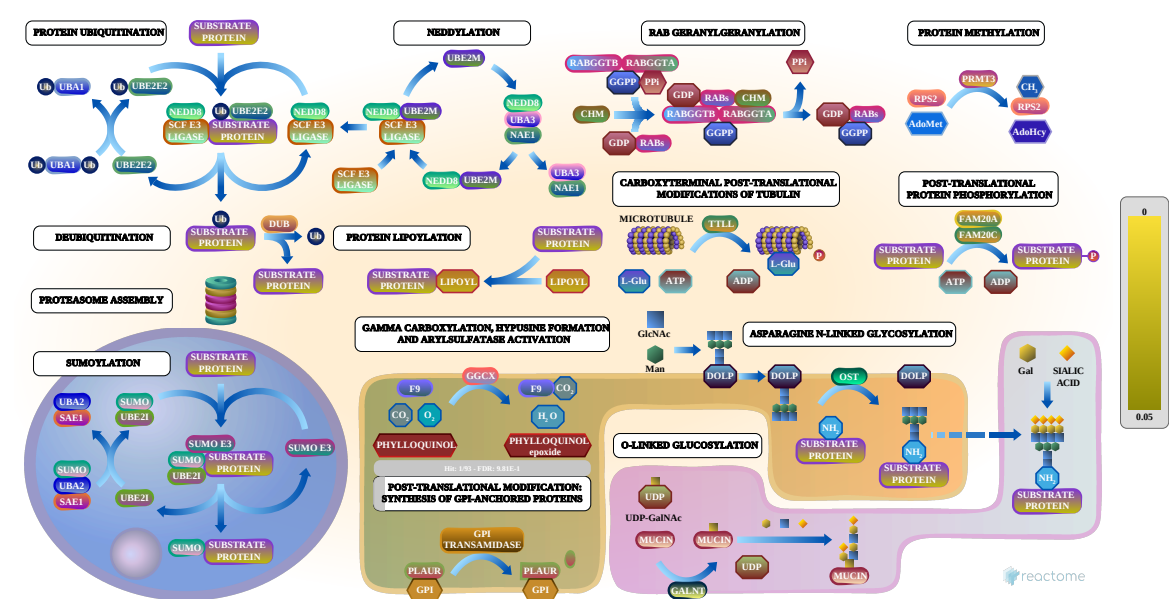
Edit history

Date	Action	Author
2008-11-23	Created	Gopinathrao G
2008-12-02	Reviewed	Jassal B, Wright EM, Matthews L
2014-06-09	Revised	Jassal B
2023-10-26	Reviewed	D'Eustachio P
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P11798	Q99624

6. Post-translational protein modification (R-HSA-597592)



After translation, many newly formed proteins undergo further covalent modifications that alter their functional properties. Modifications associated with protein localization include the attachment of oligosaccharide moieties to membrane-bound and secreted proteins (**N-linked** and **O-linked glycosylation**), the attachment of lipid (**RAB geranylgeranylation**) or glycolipid moieties (**GPI-anchored proteins**) that anchor proteins to cellular membranes, and the vitamin K-dependent attachment of carboxyl groups to glutamate residues. Modifications associated with functions of specific proteins include **gamma carboxylation** of clotting factors, **hypusine formation** on eukaryotic translation initiation factor 5A, conversion of a cysteine residue to formylglycine (**arylsulfatase activation**), methylation of lysine and arginine residues on non-histone proteins (**protein methylation**), **protein phosphorylation** by secretory pathway kinases, and **carboxyterminal modifications of tubulin** involving the addition of polyglutamate chains.

Protein ubiquitination and **deubiquitination** play a major role in regulating protein stability and, together with **SUMOylation** and **neddylolation**, can modulate protein function as well.

References

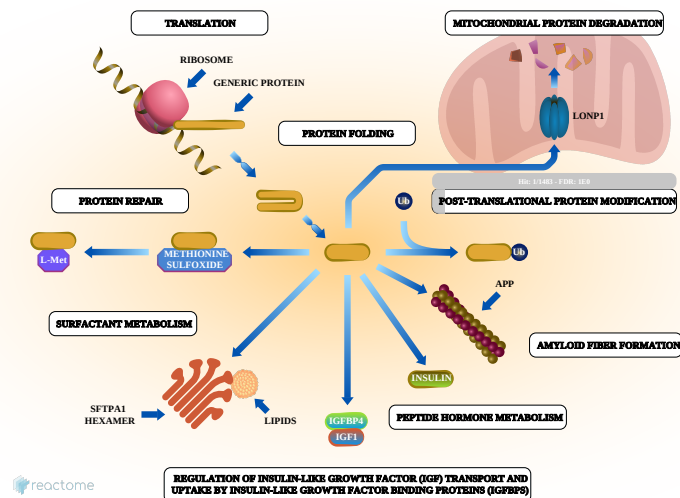
Edit history

Date	Action	Author
2005-04-18	Authored	D'Eustachio P
2010-04-14	Created	D'Eustachio P
2024-03-05	Reviewed	Matthews L
2024-08-20	Reviewed	Matthews L
2025-08-26	Edited	D'Eustachio P
2025-08-26	Reviewed	Stafford DW, Orlean P
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P26645	Q93070

7. Metabolism of proteins (R-HSA-392499)



Metabolism of proteins, as annotated here, covers the full life cycle of a protein from its synthesis to its posttranslational modification and degradation, at various levels of specificity. Protein synthesis is accomplished through the process of Translation of an mRNA sequence into a polypeptide chain. Protein folding is achieved through the function of molecular chaperones which recognize and associate with proteins in their non-native state and facilitate their folding by stabilizing the conformation of productive folding intermediates (Young et al. 2004). Following translation, many newly formed proteins undergo Post-translational protein modification, essentially irreversible covalent modifications critical for their mature locations and functions (Knorre et al. 2009), including gamma carboxylation, synthesis of GPI-anchored proteins, asparagine N-linked glycosylation, O-glycosylation, SUMOylation, ubiquitination, deubiquitination, RAB geranylgeranylation, methylation, carboxyterminal post-translational modifications, neddylation, and phosphorylation. Peptide hormones are synthesized as parts of larger precursor proteins whose cleavage in the secretory system (endoplasmic reticulum, Golgi apparatus, secretory granules) is annotated in Peptide hormone metabolism. After secretion, peptide hormones are modified and degraded by extracellular proteases (Chertow, 1981 PMID:6117463). Protein repair enables the reversal of damage to some amino acid side chains caused by reactive oxygen species. Pulmonary surfactants are lipids and proteins that are secreted by the alveolar cells of the lung that decrease surface tension at the air/liquid interface within the alveoli to maintain the stability of pulmonary tissue (Agassandian and Mallampalli 2013). Nuclear regulation, transport, metabolism, reutilization, and degradation of surfactant are described in the Surfactant metabolism pathway. Amyloid fiber formation, the accumulation of mostly extracellular deposits of fibrillar proteins, is associated with tissue damage observed in numerous diseases including late phase heart failure (cardiomyopathy) and neurodegenerative diseases such as Alzheimer's, Parkinson's, and Huntington's.

References

Edit history

Date	Action	Author
2009-03-04	Edited	Matthews L

Date	Action	Author
2009-03-05	Authored	Matthews L
2009-03-05	Created	Matthews L
2023-11-09	Reviewed	Matthews L
2025-08-28	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
P26645	Q93070

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

127 of the submitted entities were found, mapping to 129 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
A2AN08	A2AN08	B5DFC8	B5DFC8	O08539	O08539
O08553	O08553	O08749	O08749	O88343	O88343
O88935	O88935	P04256	P04256	P04636	P04636
P05064	P05064	P05202	P05202	P06685	P06685
P06837	P06837	P08413	P08413	P09671	P09671
P10637	P10637	P10719	P10719	P10852	P10852
P10853	P10853	P10860	P10860	P10922	P10922
P11505	P11505	P11507	P11507	P11798	P11798
P12369	P12369	P12382	P12382	P13832	P13832
P14733	P14733	P15532	P15532	P16858	P16858
P18266	P18266	P19491	P19491	P19783	P19783
P21575	P21575	P22892	P22892	P25444	P25444
P26284	P26284	P26645	P26645	P29341	P29341
P31016	P31016	P35465	P35465	P42208	P42208
P43006	P43006	P46935	P46935	P48722	P48722
P48962	P48962	P50580	P50580	P51881	P51881
P56399	P56399	P56480	P56480	P60335	P60335
P61264	P61264	P61967	P61967	P62806	P62806
P62908	P62908	P63318	P63318	P68368	P68368
P68404	P68404	P68433	P68433	P70336	P70336
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P97536	P97536	P97807	P97807	P99029	P99029
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Q4FZY0	Q4FZY0	Q5U216	Q5U216	Q60931	Q60931
Q61171	Q61171	Q61207	Q61207	Q61301	Q61301
Q61361	Q61361	Q61548	Q61548	Q62261	Q62261
Q62318	Q62318	Q62813	Q62813	Q64475	Q64475
Q64478	Q64478	Q64525	Q64525	Q6AY56	Q6AY56
Q6NZJ6	Q6NZJ6	Q6PDQ2	Q6PDQ2	Q6PIC6	Q6PIC6
Q6PIE5	Q6PIE5	Q7TMY8	Q7TMY8	Q80U28	Q80U28
Q80YX1	Q80YX1	Q8BGD9	Q8BGD9	Q8BGZ4	Q8BGZ4
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Q8JZQ9	Q8JZQ9	Q8K2B3	Q8K2B3	Q8R1Q8	Q8R1Q8
Q91V14	Q91V14	Q91VR2	Q91VR2	Q91XV3	Q91XV3
Q99KI0	Q99KI0	Q99P72	Q99P72	Q9CQF3	Q9CQF3
Q9CQQ7	Q9CQQ7	Q9CZU6	Q9CZU6	Q9D051	Q9D051
Q9D8N0	Q9D8N0	Q9DB20	Q9DB20	Q9DB77	Q9DB77
Q9DC28	Q9DC28	Q9EQH3	Q9EQH3	Q9JJ54	Q9JJ54
Q9QWN8	Q9QWN8	Q9QXL1	Q9QXL1	Q9QXL2	Q9QXL2

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Q9Z2L0	Q9Z2L0				

7. Identifiers not found

These 65 identifiers were not found neither mapped to any entity in Reactome.

A1L108	A2AJI0	B3EWD4	O35274	O35501	O88737	P02088	P02089
P08553	P0C0S6	P13233	P14873	P15205	P16330	P18760	P19246
P20357	P28667	P34926	P37805	P46660	P51859	P54227	P60202
P62629	P68362	P85845	P97546	P97825	P97855	Q02563	Q148V7
Q2PFD7	Q3UHD9	Q3UNH4	Q3UR70	Q4KMM3	Q5SRX1	Q60437	Q60598
Q6A065	Q6NXX7	Q6P9K8	Q6PER3	Q6PGN3	Q6S9C5	Q6TS35	Q6ZQ58
Q7TSJ2	Q80TL4	Q8BPN8	Q8C0E2	Q8C8R3	Q8CGF6	Q8VEM8	Q9DB72
Q9EPN1	Q9ERH3	Q9JKY3	Q9JLM8	Q9JME5	Q9QXS6	Q9QYR6	Q9QYX7
Q9Z218							